

invention_oevaluat[®]r

OBJECTIVE IDEA ANALYSIS

Invention Name: Retinal Prosthesis and
Method of Manufacturing a Retinal Prosthesis

Submitted By: User-provided patent PDF

Invention ID: US 8,527,057

Date of Report: 2026-08-15

This report was generated by the inventionevaluator report-automation platform from the Invention Evaluation Engine v1.6 pipeline (evidence sufficiency architecture). It compiles established findings, analytical conclusions, and an operational audit produced by the nine-phase evaluation. Scores shown are derived executive summaries of the underlying findings — they are presentation-layer derivations, not new evidence, and do not constitute an overall patentability label. This document is not legal advice.

This is a preliminary evidence-constrained evaluation based on a public patent record and selected public sources. It is not legal advice, not a patentability opinion, and not an FTO opinion.

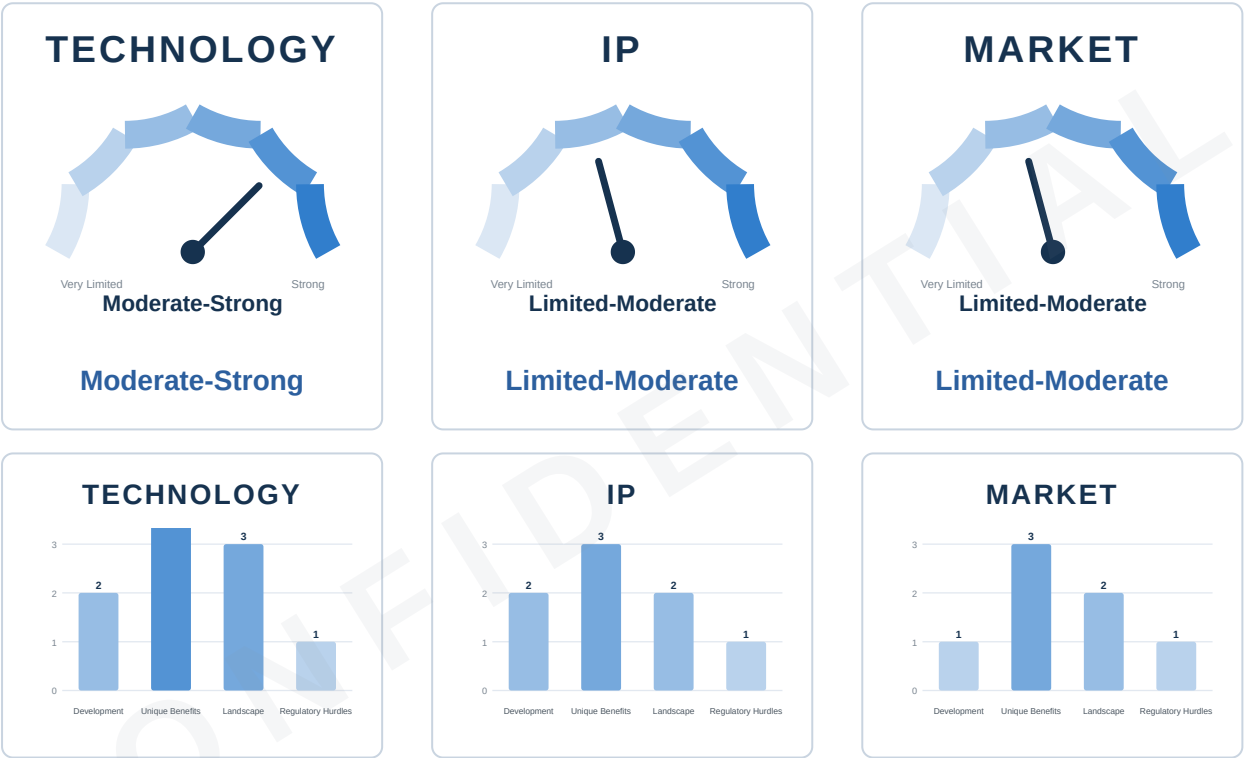
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Executive Summary

Derived executive summary — constrained to Established Findings + Analytical Conclusions (Executive Summary $\subseteq$ findings + conclusions). Gauge and bar scores are presentation-layer derivations from the findings; basis proposition_ids are listed in the scores manifest.

Very Limited Limited Limited-Moderate Moderate Moderate-Strong Strong



Commercialization Summary

Commercial actionability is Indeterminate-to-Limited on the current evidence. The patent discloses a technically differentiated implant architecture, but quantitative market size, current product embodiment, clinical performance, commercialization readiness, and current rights-chain status require additional evidence.

- High-value technical area, but high regulatory and clinical-development burden. [more](#)
- Strongest identified pathways are rights-holder diligence, specialized implant packaging, and clinical/research collaboration. [more](#)

Evidence trace

The proposition register below is extracted from the compiled report so every score basis can be checked against visible report content.

P-02-001 · P-02-002 · P-03-001 · P-03-002 · P-03-003 · P-03-004 · P-04-001 · P-04-002 · P-04-004 · P-05-001 · P-05-002 · P-06-001 · P-07-001 · P-07-002 · P-07-003 · P-08-001

CONFIDENTIAL

Technology Analysis

2.1 Technology profile

The invention is a low-profile ocular implant. An epiretinal electrode array connects through a scleral cable to an electronics package, while a secondary inductive coil receives power and data from an external coil. A molded silicone body and strap hold the elements around the sclera. The electronics package combines a ceramic substrate, conductive vias, thin-film metallization, a flip-chip ASIC, a wire-bonded passive stack, a brazed case wall, and a laser-welded lid.

2.2 Feature-benefit map

Feature	Benefit	Status
Scleral strap	Mechanical positioning	CONFIRMED PRESENT — US8527057B2 claim 1
Coplanar package/coil arrangement	Reduced implant height	CONFIRMED PRESENT — abstract/specification
Hermetic ceramic-via package	Moisture protection for implant electronics	CONFIRMED PRESENT — summary/specification
Flexible epiretinal array	Electrical access to retinal neurons	CONFIRMED PRESENT — claim 1
Suture tabs/fan tail/hook	Implant handling and fixation support	CONFIRMED PRESENT — claims 2–6

2.3 Development and regulatory status

Patent-disclosed engineering architecture; proof-of-concept, validation, clinical, production, and sales propositions remain in the Operational Audit. FDA's device overview identifies the applicable control categories; device-specific classification and pathway are not established here.

Patent Landscape Analysis

3.1 Established findings

- **P-04-001 v1:** The record sits in CPC branches covering retinal electrodes, stimulation of the eye, electromagnetic coupling, implant casings, feedthroughs, packaging, and external energy supply. Sources: Google Patents classification record; USPTO CPC A61N scheme.
- **P-04-002 v1:** The specification identifies a same-assignee design space covering retinal arrays, side-mounted coils, hermetic feedthroughs, implant packages, biocompatible bonding, and flexible circuits.

3.2 Landscape limitation

The narrowed Google Patents query A61N1/0543 AND retinal produced usable directional landscape data. Returned publication-record counts were US 492, WO 318, EP 261, CN 127, AU 105, JP 100, and CA 58. The same US query returned 492 records; the top returned applicants were Second Sight Medical Products (160), Nidek (42), Pixium Vision (32), Cochlear (28), and University of Southern California (25). These are publication-record counts and raw returned applicant names, not deduplicated patent families or normalized corporate groups.

The classification gate passed for the patent's A61N 1/18 subject matter after checking the USPTO CPC title against the retinal-prosthesis description. The chart query is the canonical narrowed query A61N1/0543 AND retinal, with jurisdiction varied per chart. No raw parent-subclass total is used as a substitute for the narrowed result. Query frequency windows are 2016–2019: 64, 2019–2022: 43, 2022–2025: 38, and 2025–2026 YTD: 5. The final bucket is partial through 2026-08-15 and is not interpreted as a decline.

IP / Novelty Analysis

4.1 Target claim

The primary target is claim 1: electrode array; sclera-surrounding strap; hermetic electronics package encasing a flip-chip circuit mounted to the strap; electrical cable; and strap-mounted coplanar secondary inductive coil powering the circuit.

4.2 Prior-art mapping

US 5,109,844 and US 5,935,155 establish retinal-array/prosthesis elements but do not establish the complete package/strap/coil combination. US 2003/0109903 establishes a related low-profile hermetic enclosure concept, but the target retinal strap and coplanar coil arrangement are not established from the reviewed record. US 7,228,181, US 7,565,203, and US 8,014,878 are related same-assignee pathways and must be treated with continuity/self-art care.

No final anticipation conclusion is admitted into Established Findings because the independent full-text and classification search was not completed across the required secondary database. This is not an FTO analysis.

4.3 Causal Bridge Test

The prior state includes retinal arrays, inductive coupling, hermetic packages, and implant fixation. The claimed state integrates these into a low-profile scleral strap with a coplanar package/coil geometry and a specified manufacturing stack. The required change is the coordinated integration of package, coil, cable, array, fixation, and hermetic manufacturing into one ocular geometry. Analogous motivation is present in the related records; direct motivation and an unexpected technical result are not established. The bridge is therefore recorded as **TRAVERSED for analytical testing**, with a partially grounded motivation and an evidentiary barrier against a stronger legal conclusion.

Literature Analysis

PubMed literature confirms the retinal-prosthesis objective, epiretinal stimulation, retinal-tack fixation, Argus II clinical context, and long-term positional-stability questions. Reviewed sources include Margalit et al. (2002, PMID 12161210), Luo and da Cruz (2016, PMID 26404104), Goetz and Palanker (2016, PMID 27502748), Ghani et al. (2023, PMID 36797684), and Palanker (2023, PMID 36781222). The literature does not establish the complete target package combination from the reviewed records.

Market Analysis

The direct market boundary is implantable retinal-prosthesis systems and their clinical support ecosystem. NEI establishes the patient need context for retinitis pigmentosa; PubMed establishes a clinical retinal-prosthesis field; FDA establishes the device regulatory controls. No reconstructable market-size, CAGR, price, reimbursement, or unit-volume figure was established. Commercial readiness remains **NOT ESTABLISHED**. For the verified adjacent manufacturing context, the 2022 Census CBP downloadable file for NAICS 334510 reports 923 establishments, 92,285 employees, and annual payroll of 10,030,531 thousand dollars. This is industry employment/payroll context, not retinal prosthesis revenue or shipment value. Employee-size establishment counts are reported in the companion chart and ledger. No unrelated NAICS code was substituted.

SWOT

Strengths: technically specific integrated implant; hermetic package detail; clear fit with a recognized retinal-prosthesis need. **Weaknesses:** performance and clinical evidence are not established from the supplied record; high development burden; current rights-chain uncertainty. **Opportunities:** rights-holder licensing/diligence; specialized implant packaging; clinical/research collaboration. **Threats:** competing retinal-prosthesis architectures; clinical and tissue risks; regulatory cost; separate FTO analysis required before product decisions.

Opportunity Assessment

Axis	Assessment	Evidence state
Market need	Medium-to-high qualitative need	CONFIRMED PRESENT — NEI/PubMed
Growth	Not established	Operational Audit
Accessibility	Low-to-medium	Analytical assessment based on FDA controls and device complexity
Competitive intensity	High	Analytical assessment; quantitative landscape incomplete
Commercial readiness	NOT ESTABLISHED	Adoption/product-identity evidence incomplete

Derived commercial actionability: **Indeterminate-to-Limited**. This is an analytical summary, not a financial forecast.

Potential Partners

The strongest identified pathways are: (1) Cortigent, the current listed assignee, for rights-chain and licensing diligence; (2) Vivani Medical for corporate assignment and historical records; (3) specialized implantable-device packaging manufacturers for co-development; and (4) ophthalmic-neuroprosthesis research and clinical centers for validation.

Publicly verified Cortigent contact: general inquiry at <https://www.cortigent.com/contact-general>; headquarters at 27200 Tourney Road, Suite 315, Valencia, CA 91355; phone 818-833-5000; Investor Relations contact Rachel Evans, investors@vivani.com. Public Vivani investor-relations pathway: <https://vivani.com/investors/>. Current willingness to license, personal decision-maker profiles, and a complete multi-company contact list remain unestablished. No partner is represented as exclusive.

Decision Matrix

Gate	Result	Premises
Utility	Analytical support at architecture level	P-03-001, P-03-002
Anticipation	Not admitted as final finding	P-05-001 remains excluded by incomplete search coverage
Obviousness bridge	Partially grounded analytical work item	P-04-002, P-05-002
Expected success	Medium, not clinically established	P-03-003
Technical effect	Low-profile integration is disclosed; performance magnitude unestablished	P-03-002, P-03-003
Commercial readiness	NOT ESTABLISHED	P-07-002

No scalar overall patentability label is produced. Any legal conclusion would require claim construction, complete prior-art coverage, prosecution history, and counsel review.

Operational Audit

Proposition	Work state	Barrier type	Remaining barrier
P-02-001 pre-filing disclosure history	EXHAUSTED	insufficient_identity	Inventor/rights-holder disclosure statement
P-03-003 performance and clinical data	EXHAUSTED	insufficient_technical_demonstration	Bench, biocompatibility, fatigue, and clinical records
P-03-004 device-specific regulatory pathway	EXHAUSTED	insufficient_identity	Product classification and intended-use record
P-04-004 quantitative landscape	SEARCHING	insufficient_cross_validation	Google Patents directional counts complete; second-source family/assignee normalization remains
P-05-001 final anticipation determination	EXHAUSTED	insufficient_search_completion	Full independent patent search and continuity review
P-05-002 stronger obviousness conclusion	EXHAUSTED	insufficient_technical_demonstration	Direct motivation, compatibility, and unexpected-result evidence
P-06-001 complete package application in literature	EXHAUSTED	source_unavailable	Engineering literature/database search

Proposition	Work state	Barrier type	Remaining barrier
P-07-001 quantitative market sizing	EXHAUSTED	source_unavailable	Reconstructable market boundary and source figures
P-07-002 commercial adoption	EXHAUSTED	insufficient_identity	Product-identity link plus two independent sources
P-08-001 partner fit	SEARCHING	insufficient_identity	Current ownership, contact, and technical-need verification

Appendices

Appendix A — Source register

1. User-provided US8527057 .pdf. 2. Google Patents, US8527057B2: <https://patents.google.com/patent/US8527057B2/en> 3. USPTO CPC A61N scheme: <https://www.uspto.gov/web/patents/classification/cpc/html/cpc-A61N.html> 4. FDA Overview of Device Regulation: <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/overview-device-regulation> 5. NEI Retinitis Pigmentosa: <https://www.nei.nih.gov/learn-about-eye-health/eye-conditions-and-diseases/retinitis-pigmentosa> 6. PubMed records: PMID 12161210, 26404104, 27502748, 36797684, 36781222.

Appendix B — Reproducible query log

- Google Patents: US8527057B2; reviewed family, classifications, continuity, and cited references.
- USPTO CPC: A61N branches for retinal electrodes, eye stimulation, electromagnetic coupling, casings, feedthroughs, packaging, and external energy supply.
- PubMed: retinal prosthesis epiretinal; reviewed 252-result search page and selected identity-qualified records.
- NEI: retinitis pigmentosa; reviewed disease burden and treatment context.
- FDA: Overview of Device Regulation; reviewed classification, IDE, QMSR, labeling, and MDR controls.

Appendix C — Original submission record

Invention: Retinal Prosthesis and Method of Manufacturing a Retinal Prosthesis.

Inventors: Robert J. Greenberg; Jerry Ok; Jordan Neysmith; Kevin Wilkins; Neil Hamilton Talbot; Da-Yu Chang. **Original assignee:** Second Sight Medical Products, Inc., Sylmar, California. **Priority:** US 60/675,980, filed 2005-04-28. **Parent:** US 11/385,315, filed 2006-03-20; issued as US 7,881,799. **Divisional:** US 12/976,650, filed 2010-12-22. **Publication:** US 2011/0118807 A1, published 2011-05-19. **Grant:** US 8,527,057 B2, granted 2013-09-03. **Classification shown on patent:** A61N 1/18; USPC 607/54.

Short description: An implantable epiretinal prosthesis combines a retinal electrode array, a hermetic electronics package, and a secondary inductive receiving coil on a sclera-

surrounding strap. The package uses ceramic vias, flip-chip ASIC attachment, wire-bonded passive components, a brazed case wall, and a laser-welded lid.

Detailed description: A flexible epiretinal array connects through a scleral cable to the package. The secondary coil receives power and data from an external primary coil. A silicone body holds the package and coil in an end-to-end or surrounding arrangement and narrows into a strap with suture tabs and an optional surgical-tool hook. The package process includes vacuum bake-out, getter material, biocompatible metals, underfill, silicone overmolding, brazing, and laser welding.

Innovation claims: Claim 1 recites the electrode array, scleral strap, hermetic flip-chip package, coupling cable, and coplanar strap-mounted secondary coil. Dependent features include suture tabs, fan tail, hook, polymer-sandwiched metal-trace cable, cable folding, wound-wire coil, and oval coil.

Proof-of-concept / commercialization: Not established from the supplied patent PDF. Human clinical testing, production, sales, and current product embodiment require separate evidence.

Disclosure history: Pre-filing public disclosure and sale-offer history are **NOT ESTABLISHED**. No inventor statement was supplied. Post-filing patent publication dates are established above. Required follow-up is a dated statement covering talks, publications, demonstrations, offers for sale, clinical disclosures, grants, and social-media disclosures.

IP posture: The record claims priority to the provisional and is a divisional of US 7,881,799. The patent states NIH grant R24EY12893-01 and government rights. Current enforceability and expiration require a separate current-status check.

Appendix D — Avenue ledger

See `avenue-ledger-us8527057-v16.md` for the complete work-state and escalation record.

SWOT Analysis

STRENGTHS

- Detailed integrated implant architecture (P-03-001)
- Hermetic package and low-profile arrangement are technically specific (P-03-002)
- Clear fit with retinal-prosthesis need and research field (P-07-003)

WEAKNESSES

- Clinical and comparative performance not established (P-03-003)
- Quantitative market size not established (P-07-001)
- Current rights-chain and product embodiment require verification (P-02-002/P-07-002)

OPPORTUNITIES

- Rights-holder diligence and licensing pathway (P-08-001)
- Hermetic implant packaging collaboration
- Clinical/research partnerships around implant stability and retinal stimulation

THREATS

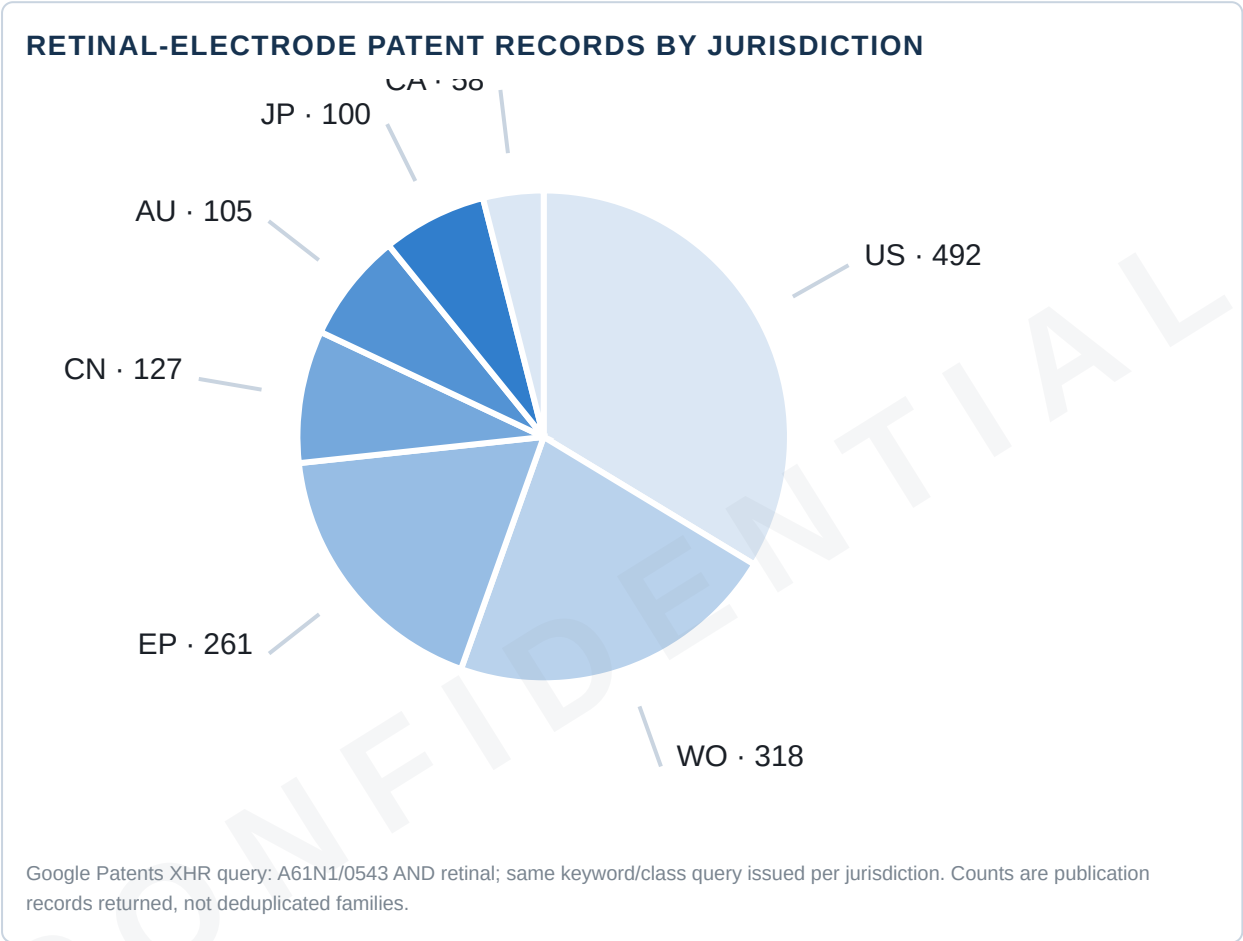
- High regulatory and clinical burden (P-03-004)
- Competing epiretinal, subretinal, photovoltaic, and cortical architectures
- FTO and current legal-status questions require separate counsel-led work

COMMERCIAL
ACTIONABILITY

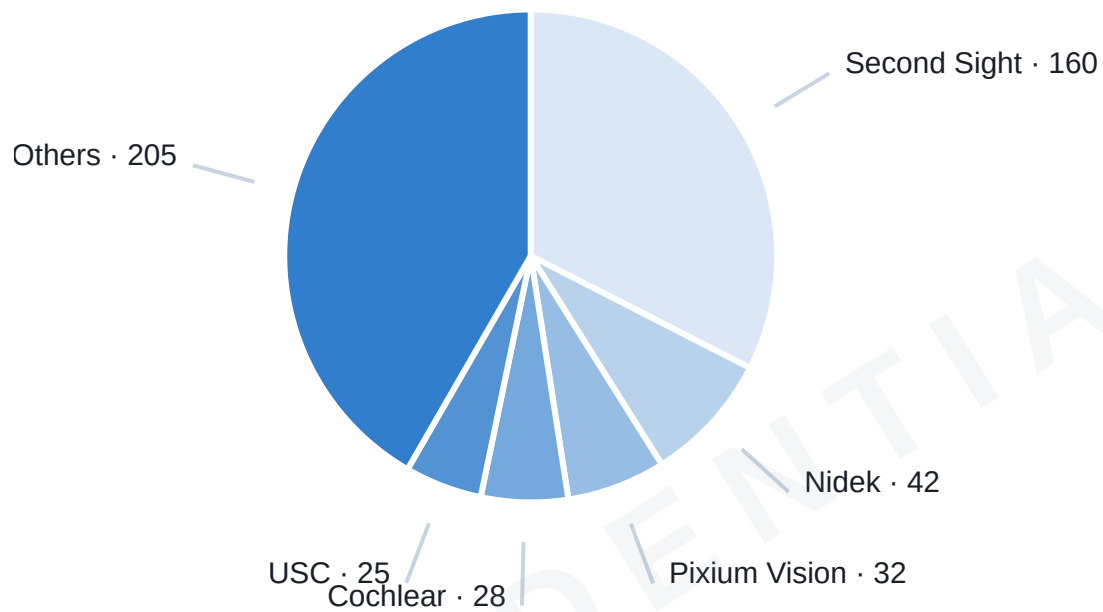
Indeterminate-to-
Limited

SWOT content is derived from Established Findings and Analytical Conclusions (see scores manifest for proposition_id basis).

Landscape & Market Data

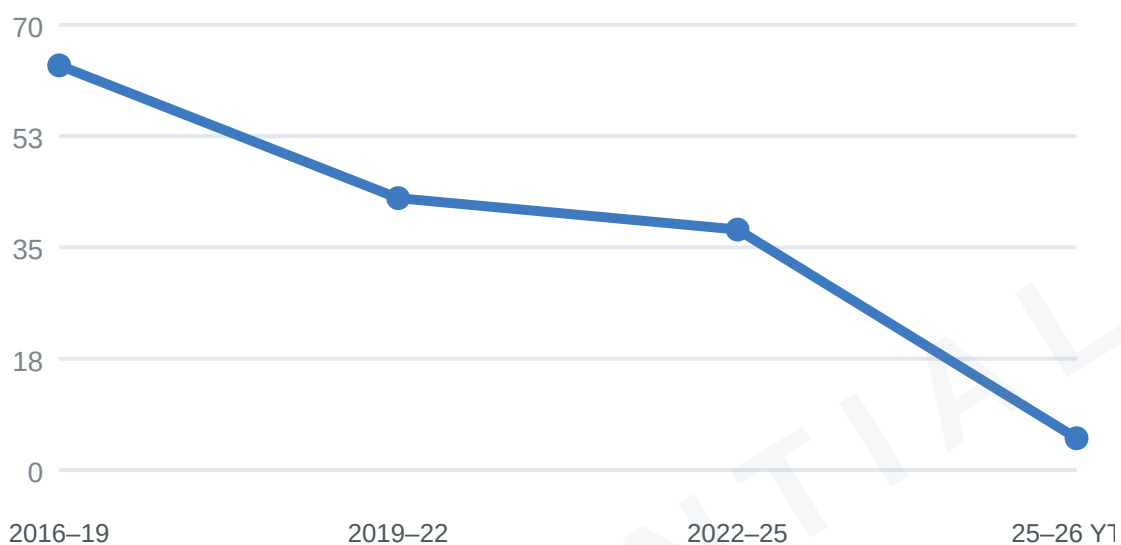


TOP APPLICANTS IN THE RETINAL-ELECTRODE QUERY



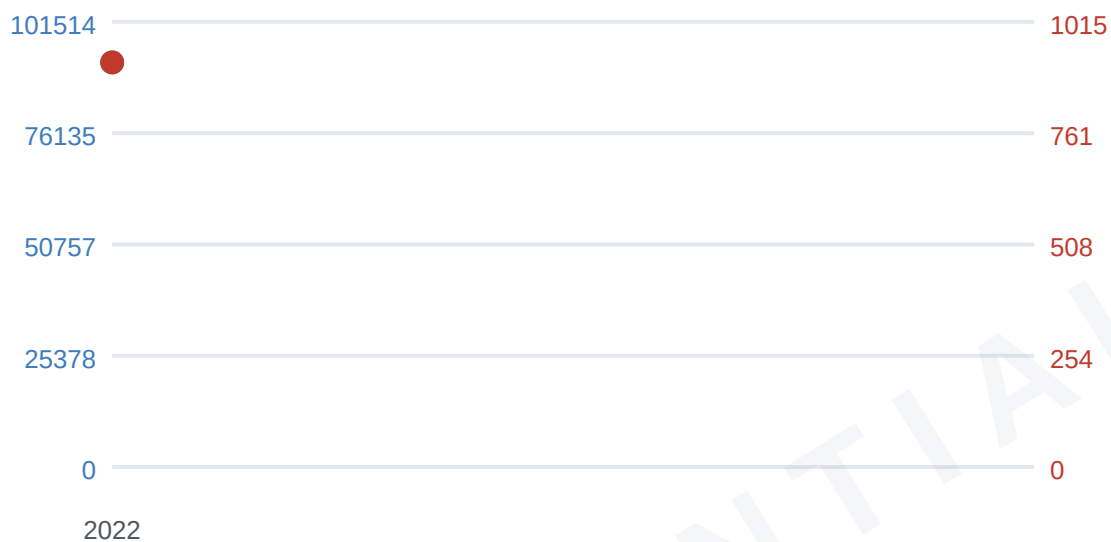
Google Patents XHR query: country=US; A61N1/0543 AND retinal; 492 results. Display labels shorten returned applicant names; names have not been merged across corporate variants.

RETINAL-ELECTRODE FILING ACTIVITY WINDOWS



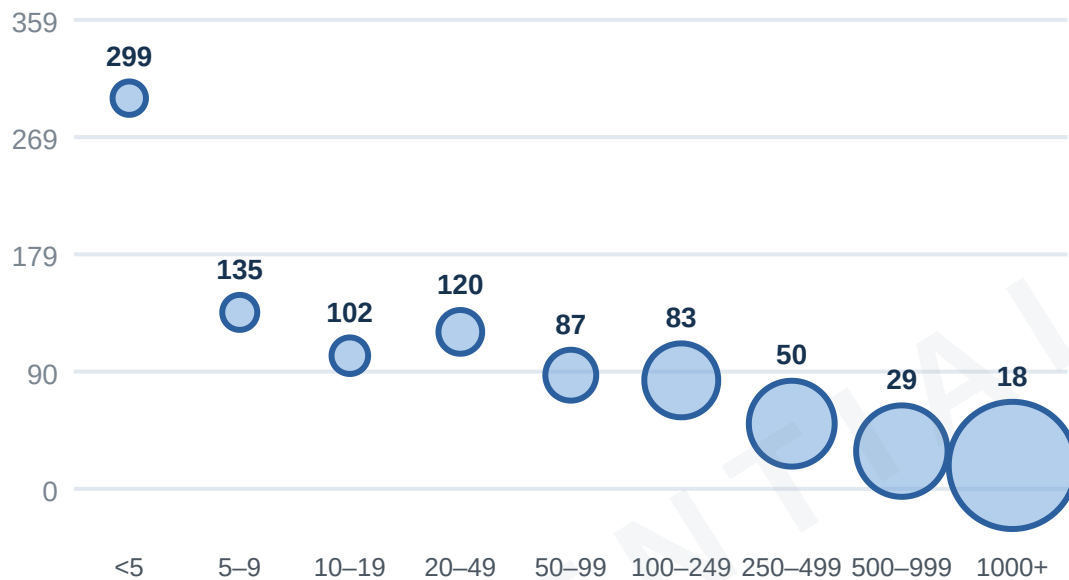
Google Patents query frequency buckets, not a deduplicated annual filing series; 2025-2026 YTD is partial through 2026-08-15 and is not a decline signal.

NAICS 334510 — EMPLOYEES VS ESTABLISHMENTS, 2022



U.S. Census Bureau CBP 2022 downloadable file cbp22us.zip; NAICS 334510, lfo='-' (all legal forms). Employees and establishments are reported values.

NAICS 334510 ESTABLISHMENTS BY EMPLOYEE-SIZE BRACKET, 2022



U.S. Census Bureau CBP 2022 downloadable file cbp22us.zip; n-fields are establishment counts and e-fields are employee counts.